Introduction to medical images: Ultrasonography

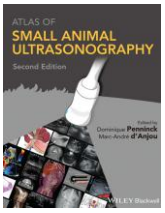
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National Taiwan University Veterinary Hospital

What is ultrasonography?

- Diagnostic medical procedure
- Use sound waves to produce Black/white image on screen
- View internal structures



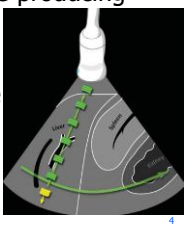
How it works?

- Transducer (medical imaging):3-12 MHz
- Sound waves are either absorbed or bounce back to crystals in the head of the transducer



nationalultrasound.com

- Image is formed by hundreds of individual scan lines
- Echoes (green) generated at interfaces of different acoustic impedance producing variable echogenicity
- Each echo is anatomically localized based on the time interval between emitted pulse and its reception.



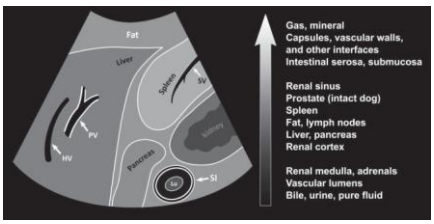
Echogenicity

- Hyperechoic
  - ✓High reflection
- Hypoechoic
  - ✓Intermediate reflection
- Anechoic
  - ✓No reflection

Table 1.1
Density and speed of sound in materials and biological tissues and resulting acoustic impedance

Material or tissue	Density (kg/m^3)	Speed (m/s)	Acoustic impedance
Air	1.2	330	0.0004×10^6
Lung	300	600	0.18×10^6
Fat	924	1,450	1.34×10^6
Water	1,000	1,480	1.48×10^6
Soft tissues (in general)	1,050	1,540	1.62×10^6
Liver	1,061	1,555	1.65×10^6
Kidney	1,041	1,565	1.63×10^6
Skull bone	1,912	4,080	7.8×10^6

Source: Bushberg et al. (2011).



Transducers

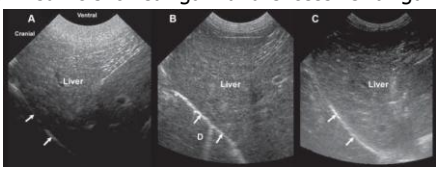
- Convex (A, B) and linear (C)
- A: lower frequency, large dog, abdomen
- B: higher frequency,small patient, abdomen
- C: superficial structures on wide/flat surface
- D: echocardiography



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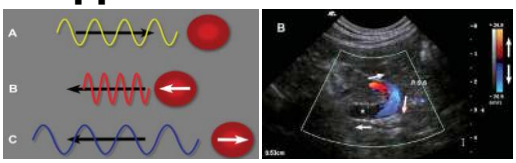
Gain setting

- Ultrasound attenuate when travel through tissue
- Amplification of echoes received must be adjusted
- A: excessive near gain and insufficient far gain
- B: well-adjusted near and far gains
- C: insufficient near gain and excessive far gain



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Doppler effect

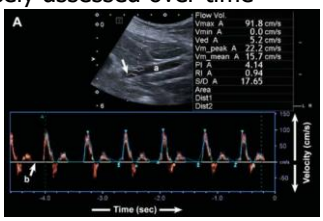


- B: RBC moves toward this pulse, increasing the frequency of the returning echo. (RED)
- C: RBC moves away from this pulse, the frequency of the returning echo is reduced. (BLUE)

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Spectral (pulsed-wave) Doppler

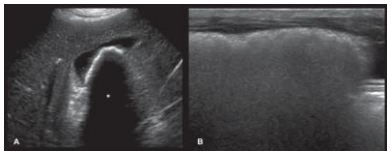
- Examines blood flow at a specific site
- velocity, direction and uniformity can be precisely assessed over time



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Common artifacts

- Acoustic shadowing
  - ✓Zone of echoes with reduced amplitude beyond a highly attenuating or reflective structure

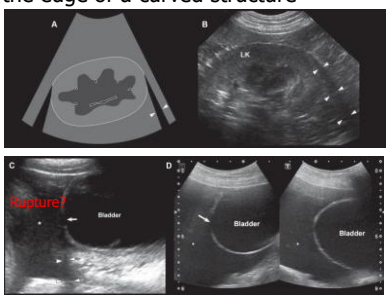


A: Clean shadowing, such as bone, calculi,

B: Dirty shadowing, mixed gas and stools present in the colon

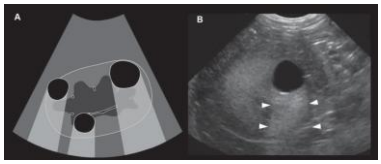
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- Edge shadowing
  - ✓Discrete, triangular zones of low amplitude, at the edge of a curved structure



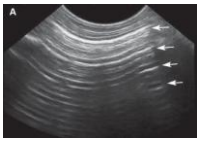
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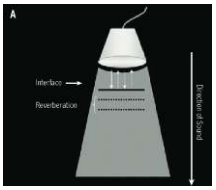
- Acoustic enhancement
  - ✓Waves pass through easily, remain higher intensity when reaching the deeper tissues =>allowing greater echoes return to the probe =>deeper tissues present an increased artifact in echogenicity



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- Reverberation
  - ✓Series of parallel and equally spaced lines (arrows), when the beam encounters a highly reflective interface such as gas.

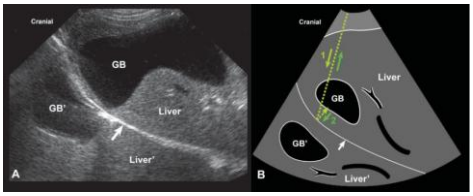




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Rehabilitative Ultrasound Imaging: Understanding the Technology and Its Applications

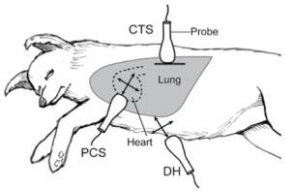
- Mirror image
  - ✓When a large, smooth, curvilinear, and strongly reflective interface between tissues



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Thorax Ultrasonography

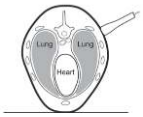
- Chest tube site, CTS
- Pericardial site, PCS
- Diaphragmatico-hepatic, DH

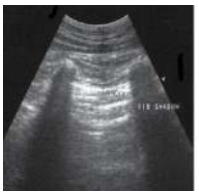


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Gator (alligator) sign

- Chest tube site
  - ✓Pleural pulmonary line, PP-line



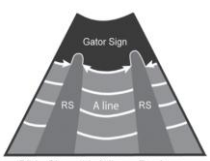




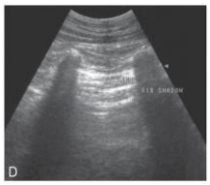
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Normal dry lung

- Glide sign
  - Motion of PP-line with respiration
- A-line



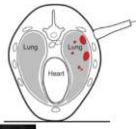
C Glide Sign with A-lines, Dry Lung

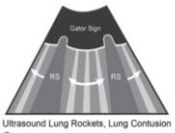

D

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Wet lung

- Ultrasound Lung Rocket (B-line)
  - Wedge-shaped
  - Arise from the pleural line
  - Extending through the far field
  - Move with the glide sign





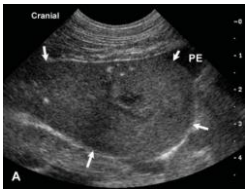
Ultrasound Lung Rockets, Lung Contusion
C


D

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Lung lobe torsion

- Lung lobe appears "hepatized" and enlarged (arrows).



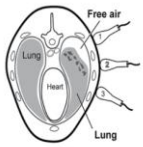
A


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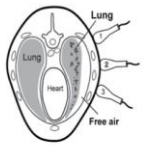
4 year old Afghan Hound (veterinaryradiology.net)

Pneumothorax

- Presence of A-lines without a glide sign
- Rule out by the presence of glide sign/ ultrasound Lung Rocket



B Partial PTX

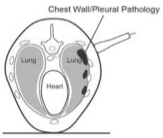


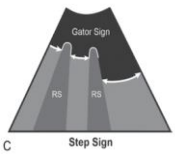
C Massive PTX

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Chest wall/Pleural pathology

- Step sign
  - Disruption of normal linear continuity of PP-line





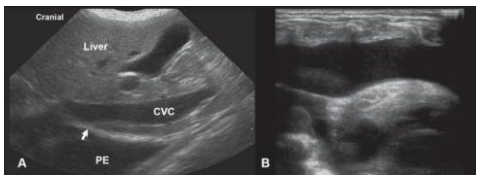
C Step Sign


D

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Pleural effusion

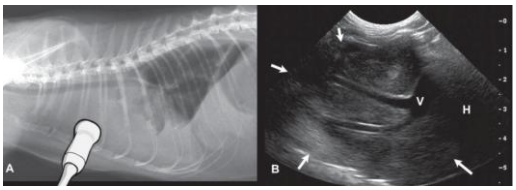
- (A) Subcostal/transhepatic approach
- (B) Intercostal approach



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Mediastinum

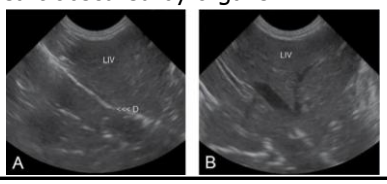
- Differentiating fat accumulation from fluid or mediastinal masses



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Diaphragmatic herniation

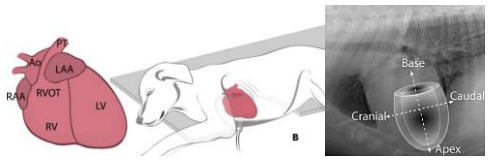
- Diaphragmatico-hepatic, DH
  - ✓ Loss of hyperechoic curvilinear continuity of diaphragm
  - ✓ Location of liver, GB
  - ✓ Heart obscured by organs



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Heart Ultrasonography (Echocardiography)

Orientation and anatomy





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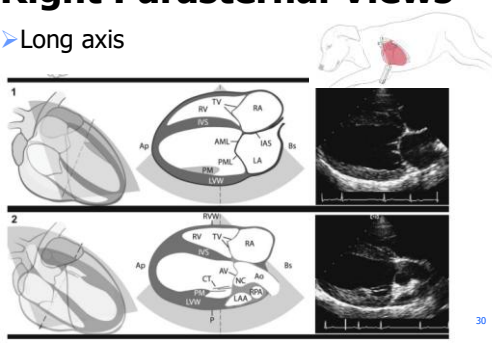
Evaluate

- Chamber
- Muscle
- Valve
- Stenosis
- Systolic/diastolic function
- Velocity
- Regurgitation

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Right Parasternal Views

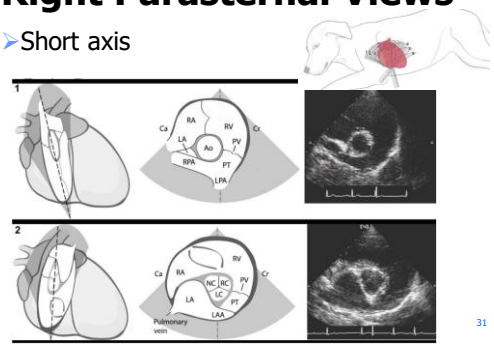
- Long axis



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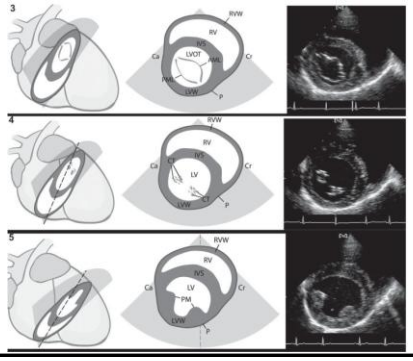
Right Parasternal Views

- Short axis



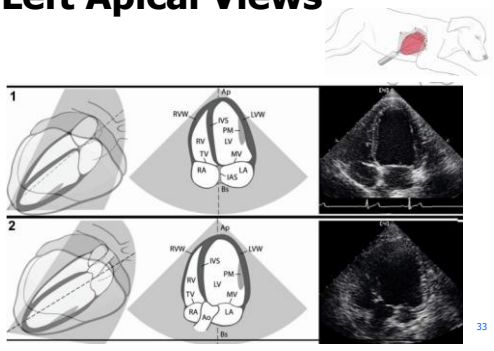
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Right Parasternal Views



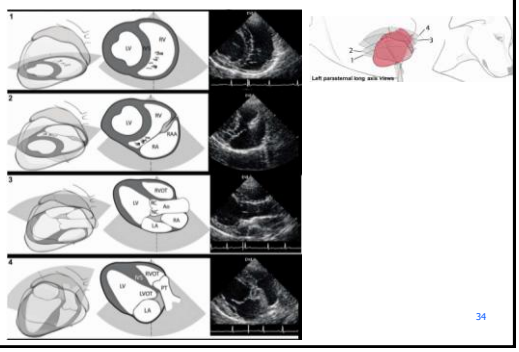
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Left Apical Views



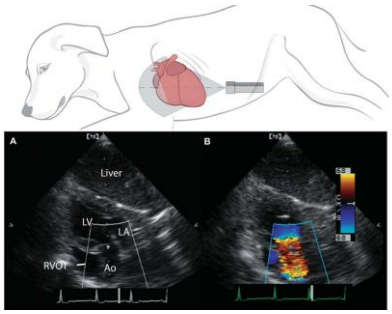
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Left Parasternal Views



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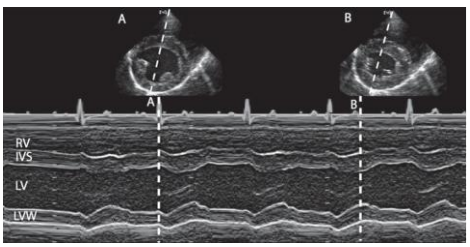
Subcostal Views



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Motion-mode Echocardiography

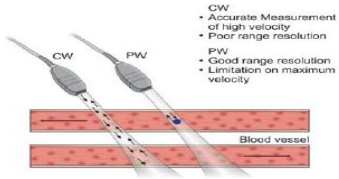
- Recording of time-dependent short-axis dimensions of the heart



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Doppler Echocardiography

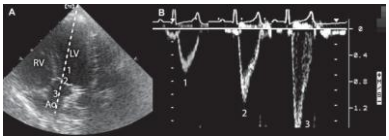
- Pulsed-wave Doppler (PWD)
  - Determine the velocity of the specific point
- Continuous-wave Doppler (CWD)
  - No practical velocity limit



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Velocity

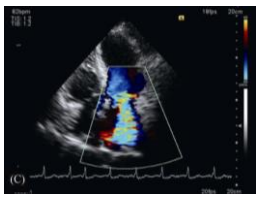
- Baseline: 0 m/s
- Upper: toward to transducer
- Lower: away from transducer
- Parallel to blood flow as possible

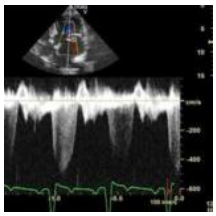


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Chronic degenerative valve disease (CDVD)

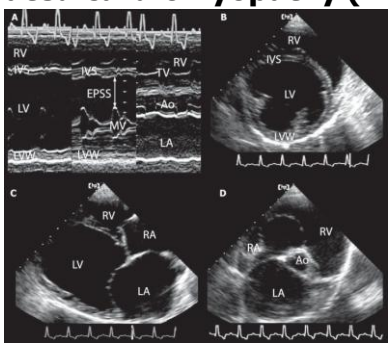
- Mitral valve regurgitation





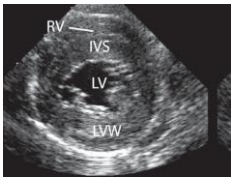
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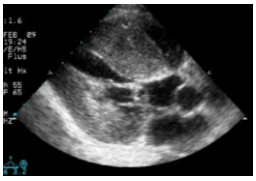
Dilated cardiomyopathy (DCM)



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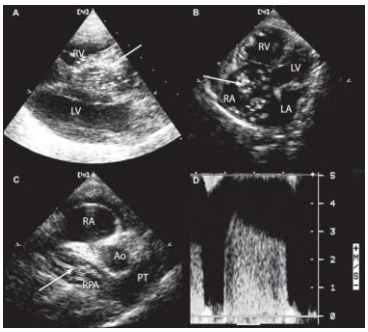
Hypertrophic cardiomyopathy (HCM)





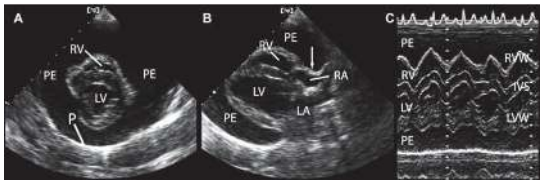
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Heartworm (dirofilariasis)



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Pericardial effusion and tamponade

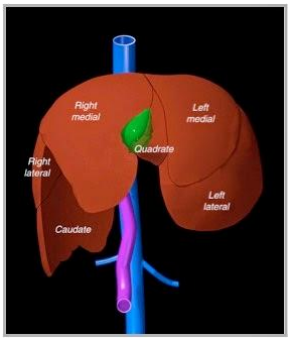


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Abdomen Ultrasonography

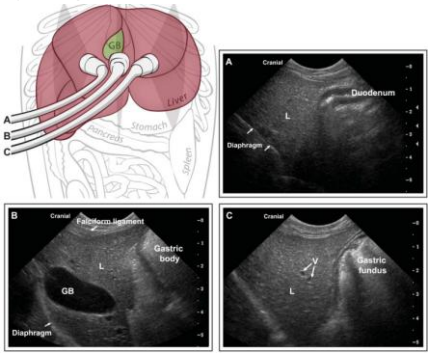
Liver

- 6 lobes
  - ✓Caudate
  - ✓Right lateral
  - ✓Right medial
  - ✓Quadrate
  - ✓Left lateral
  - ✓Left medial



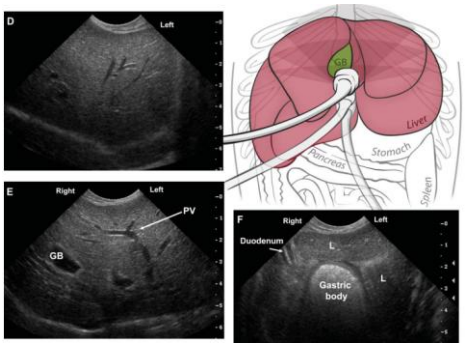
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- Sagittal plans



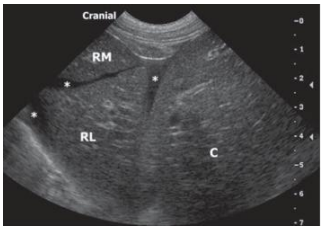
46

- Transverse plans



47

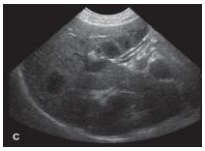
- Effusion(*) separating hepatic lobes. C, caudate lobe; RL, right lateral; RM, right medial.



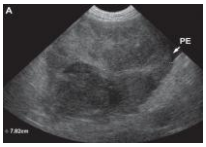
48

- Hepatic neoplasm





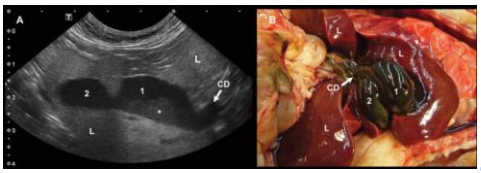




49

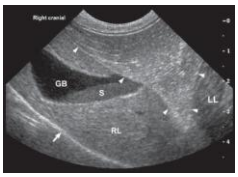
Biliary System

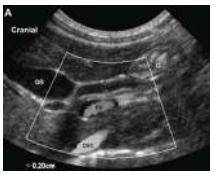
- Gallbladder
  - ✓Anechoic teardrop-shaped structure
  - ✓Occasionally bilobed in cats
  - ✓With a cystic duct



50

- GB is thin and smooth, less than 1 mm
- Sludge often recognized without signs
- Movement of sludge should respond to gravity after animal repositioned



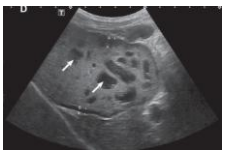


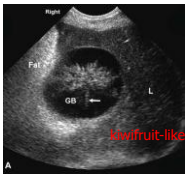
51

- Gallbladder wall edema and mucocele







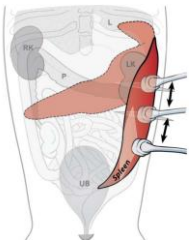


kiwifruit-like pattern

52

Spleen

- Left craniodorsal abdomen
- Caudal to stomach
- Adjacent to body wall
- Position varies with gastric filling



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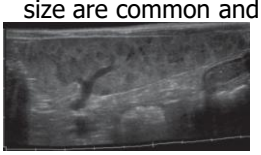
Liver, spleen, kidney relative echogenicities and echotextures

- Liver is relatively hypoechoic and more granular in echotexture
- Liver is relatively isoechoic to the cortex of the right kidney.

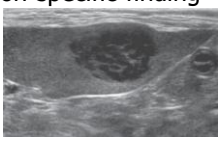


54

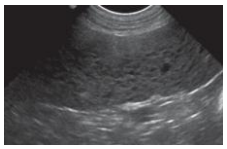
- Nodules of variable echogenicity and size are common and non-specific finding



Mutifocal splenic infarctions and necrosis



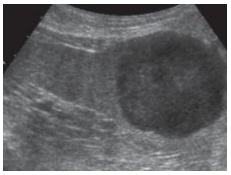
Fibrous histiocytic nodule



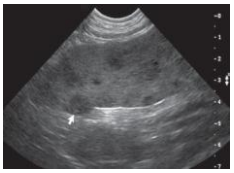
Splenic lymphoma



Extramedullary hematopoiesis



Pyogranulomatous splenitis



Disseminated histiocytic sarcoma

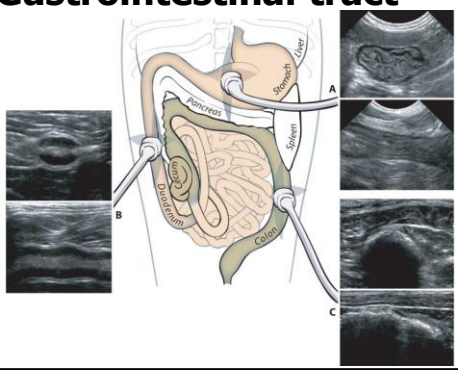


Splenic myelolipomas

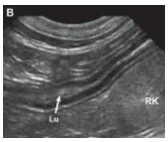


Hemangiosarcoma

Gastrointestinal tract



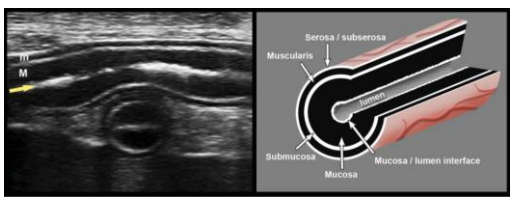




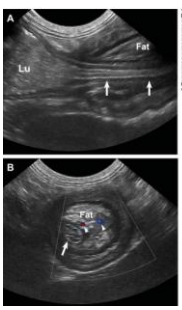


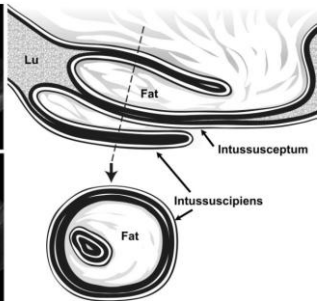
	Stomach	Duodenum	Jejunum	Ileum	Cecum/Colon
Dog					
<15 kg	2–5 mm	3.8	3.0	3.0	1.5
15–30 kg	2–5 mm	4.1	3.5	3.5	1.5
>30 kg	2–5 mm	4.4	3.8	3.8	1.5
Cat	2–4 mm	2.2	2.2	2.8	1.5

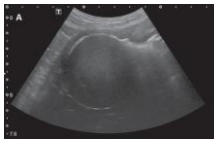
- Normal intestinal layers



- Intussusception







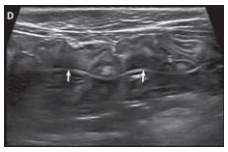
Gastric foreign bodies: play balls



Elastic bands



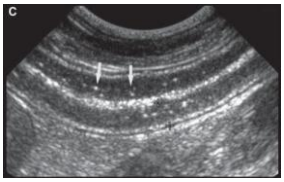
hard plastic foreign body



Linear foreign bodies

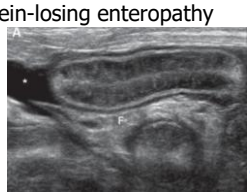
61

- Increased mucosal echogenicity and mucosal speckles
  - ✓Lymphocytic-plasmacytic enteritis
  - ✓Recently fed; speckles disappear at the end of digestion



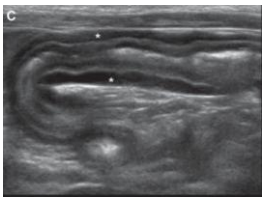
62

- Mucosal hyperechoic striations
  - ✓Linear hyperechoic lines within the mucosa and aligned perpendicularly to the lumen axis
  - ✓Represent dilated lacteals
  - ✓Associated with protein-losing enteropathy and lymphangiectasia



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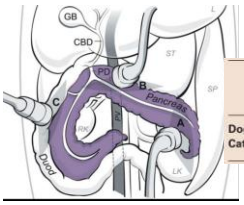
- Intestinal wall muscular thickening
  - ✓Mechanical obstruction secondary to foreign material
  - ✓Inflammatory bowel disease
  - ✓Tumor infiltration



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Pancreas

- Left (A) and right lobes (C) and body (B)

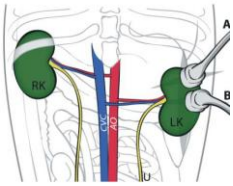


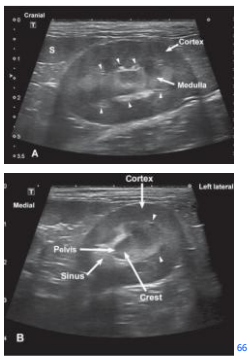
	Mean left lobe thickness	Mean body thickness	Mean right lobe thickness	Pancreatic duct diameter
Dogs[a]	6.5 mm	6.3 mm	8.1 mm	0.6 mm
Cats[b]	5.8 mm	6.2 mm	4.4 mm	1 mm[c]

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Kidney

- Longitudinal (A)
- Transverse (B)

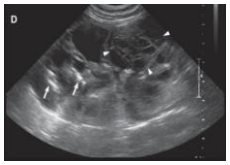




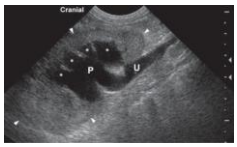
66



Chronic renal disease



polycystic kidney



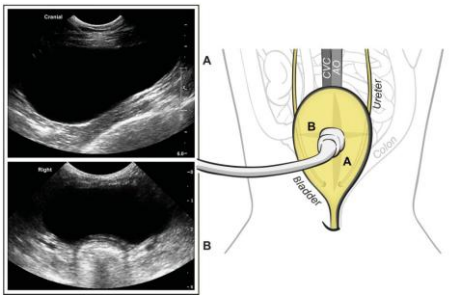
Obstructive hydronephrosis.



Hydroureter

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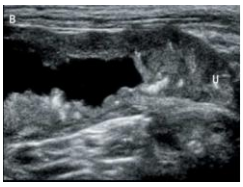
Bladder



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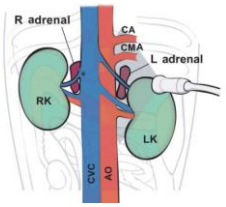
Cystitis



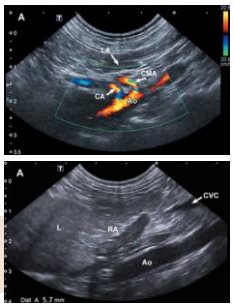
Transitional cell carcinoma (TCC)

69

Adrenal gland



	Thickness or width (mm)[a]	Length (mm)[b]
Dogs	6–8	10–50
Cats	3.5–4.5	10–11



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Thank you for your attention!

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